

MINA ABDEL-MASIH

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Education

North Carolina State University

Expected May 2028

Bachelor of Science in Computer Science (AI Concentration), Math Minor | GPA: 4.0/4.0

Raleigh, NC

- Ranked **#1** in Computer Science (**306**), Engineering (**2,205**), and Junior Class (**7,793**)
- Honors/Awards:** NCRMA Scholar, Henry B Rowe Jr. Scholar, Robert F. & Romaine S. Angel Scholar, Dean's List
- Courses:** Data Structures & Algorithms, Automated Learning & Data Analysis, C & Software Tools, Software Development, Operating Systems, Information Systems Management, Intro to Artificial Intelligence, Models of Computation & Languages

Work Experience

Zaki Technology

May 2026 – August 2026

Automation Developer Intern

Durham, NC

- Built an **n8n** pipeline integrating **Gmail** and the **Gemini API** to auto-classify daily insurance loss alerts
- Automated a manual **2–3 hour** daily review process, classifying **20–30 alerts/day** by loss type and duplicate status
- Designed a Gmail-to-Sheets architecture, replacing manual spreadsheet entry with structured **JSON**-based data logging

North Carolina State University

Aug. 2024 – Dec. 2024

Information Security Research

Raleigh, NC

- Evaluated **6** peer-reviewed authentication studies, demonstrating how **2FA** cuts account compromise risk by **≈99%**
- Transformed a **6**-source literature review into a responsive, accessible site with scrollspy nav
- Engineered a **React + Vite** rewrite: **17** scoped components, **6** hooks, and a DPR-capped canvas engine

Walgreens

Oct. 2022 – Present

Pharmacy Customer Service Associate

Raleigh, NC

- Enrolled **100-200** customers in Walgreens credit card program through customer-first service
- Earned Employee of the Month multiple times for consistent service quality and reliability
- Processed patient and drug data in pharmacy systems under HIPAA and state/federal compliance standards

Projects

Rooted | *Two-Sided Therapy Journaling App* | *Python, FastAPI, watsonx, PostgreSQL*

March 2026

- Built a **4**-tool **LangChain/watsonx** agent that branches on a local safety verdict, skipping the LLM on crisis paths
- Engineered a **3**-tier fallback (**Orchestrate** → **granite-13b** → **Python**) so nothing fails when the model is down
- Architected an async **FastAPI/PostgreSQL** API with **12 JWT**-secured endpoints and role-scoped content redaction
- Developed role-routed patient and clinician UIs in **Streamlit**, containerized via **Docker Compose** for one-command setup

Student Productivity Tool | *Spring Boot, React, PostgreSQL*

May 2026

- Architected a full-stack academic tracker with **20** REST endpoints on **Spring Boot, React, and PostgreSQL**
- Engineered a stateless engine computing weighted grades, target-score projections, and credit-weighted GPA
- Developed an **87**-test **JUnit** suite reaching **100%** line coverage across all domain and calculation classes

Access Control Compliance Engine | *Java, JUnit 5*

February 2026

- Engineered a **Java** access-control auditor that flags **262K** violations across **126K** users from **2M+** records in **35s**
- Benchmarked **9** pluggable map implementations behind one factory; **skip-list** swap cut runtime **66x** at **65K** records
- Drove the full **SDLC**: proposal, **UML** design, black-box test plan, and **JUnit 5** regression suites

MinaAbdel-Masih.com | *Portfolio Website* | *React, WebGL*

August 2026

- Engineered a serverless **Gemini** proxy with server-side keys, per-IP rate limiting, and prompt-injection guards
- Optimized initial page load **83%** (1,009 KB → 171 KB) by lazy-loading the **WebGL** skills bundle
- Built a **4**-module data layer driving **13 React** components and the AI system prompt from one source of truth

Technical Skills

Programming Languages (Proficiency): Java, C, Python, LaTeX, HTML/CSS, JavaScript

Technologies: Git, GitHub, Microsoft Office, VSCode, Eclipse, Google Workspace, React.js, Claude Code, Postman, n8n, Jenkins, REST API, FastAPI, JUnit

Certifications: ChatGPT Prompt Engineering for Developers, IBM Generative AI Essentials, Wells Fargo Software Engineering Job Simulation Certificate, IBM RAG for Enhanced AI Output

Student Organizations & Extracurriculars

Artificial Intelligence Club | NCSU App Dev Club | NCSU Oriental Orthodox Club (*Treasurer*) | UNC Coptic Student Organization | NCSU Arab Student Association | UNC Arab Student Association | NCSU Society of Sales Engineers